

Exercise I.

[20 marks]

Compute the following integrals:

1. $\int x^5 \ln(-x) dx,$

3. $\int \frac{\tan x \sin x}{\sin^3 x + \cos^3 x} dx,$

2. $\int \frac{x^5 + 2x^3 + x^2 + 3}{x^2 + 2} dx,$

4. $\int \frac{x}{(1+x^2)\sqrt{1+x^2}} dx.$

Exercise II.

[20 marks]

Let $\alpha \in \mathbb{R}$. Study the natures of the following series:

1. $\sum_{n=2}^{\infty} \frac{\ln n}{n^4},$

3. $\sum_{n=2}^{\infty} (-1)^n \cos n,$

2. $\sum_{n=2}^{\infty} \frac{\alpha^n}{\ln n},$

4. $\sum_{n=2}^{\infty} \frac{(-1)^n}{n \ln n} \cos \frac{1}{n}.$

Exercise III.

[20 marks]

Let $\alpha \in \mathbb{R}$. Study the natures of the following improper integrals:

1. $\int_0^1 \frac{1 - \cos x}{x^\alpha \sqrt{1-x}} dx,$

3. $\int_1^{\infty} \sin(x^3 + 1) dx,$

2. $\int_1^{\infty} \frac{1}{e^x + e^{-x}} dx,$

4. $\int_0^{\infty} \frac{\sin x}{x^{\frac{3}{2}}} dx.$

Exercise IV.

[30 marks]

1. Calculate the following limit:

$$\lim_{n \rightarrow \infty} \left(\prod_{k=1}^n e^{\sqrt{\frac{n^2 - k^2}{n^2}}} \right)^{\frac{1}{n}}.$$

2. Let $g(x) = \int_0^{\cos x} [\ln(1-t^2)]^2 dt$. Compute $g'(x)$ and specify the domain of differentiability of g .

3. Let $f(t) = \begin{cases} \frac{\sin t}{t} & \text{if } t \neq 0 \\ 1 & \text{if } t = 0. \end{cases}$ Calculate the following limit:

$$\lim_{x \rightarrow 1} \frac{1}{x-1} \int_0^{\ln x} f(t) dt.$$

Exercise V.

[10 marks]

Let $I_n = \int_0^1 x(\ln x)^n dx$.

1. Show that the improper integral I_n is convergent for all $n \in \mathbb{N}$.

2. Show that $I_{n+1} = -\frac{n+1}{2} I_n$ and deduce the value of I_n .

Exercise 1. Compute the following integrals:

$$1. \int x^5 \ln(-x) dx = \frac{x^6}{6} \ln(-x) - \int \frac{x^6}{6} \cdot \frac{1}{x} dx$$

$$= \frac{x^6}{6} \ln(-x) - \frac{1}{6} \int x^5 dx = \frac{x^6}{6} \ln(-x) - \frac{x^6}{6^2} + C$$

① $u = \ln(-x)$ $v' = x^5$
 ② $u' = \frac{1}{x}$ $v = \frac{x^6}{6}$

$$2. \int \frac{x^5 + 2x^3 + x^2 + 3}{x^2 + 2} dx = \int \left[x^3 + 1 + \frac{1}{x^2 + 2} \right] dx$$

$$= \frac{x^4}{4} + x + \frac{1}{\sqrt{2}} \arctan \frac{x}{\sqrt{2}} + C$$

③

$x^5 + 2x^3 + x^2 + 3$ $\ominus x^5 + 2x^3$	$x^2 + 2$ $x^3 + 1$
$x^2 + 3$ $\ominus x^2 + 2$	1

$$3. \int \frac{\tan x \sinh x}{\sinh^3 x + \cosh^3 x} dx = \int \frac{\tan x}{\frac{\sinh^3 x + \cosh^3 x}{\sinh x \cosh^2 x}} \frac{dx}{\cosh^2 x}$$

$$= \int \frac{t}{t^2 + \frac{1}{t}} dt = \frac{1}{3} \int \frac{3t^2}{t^3 + 1} dt$$

$$\stackrel{①}{=} \frac{1}{3} \ln |t^3 + 1| + C = \frac{1}{3} \ln |\tan^3 x + 1| + C$$

Bigche Rule:
 $\omega(\pi + x) = \omega(x)$
 $t = \tan x = u(x)$ ②
 $dt = \frac{1}{\cos^2 x} dx$

$$4. \int \frac{x}{(1+x^2)\sqrt{1+x^2}} dx = \int \frac{\tan t}{(1+\tan^2 t)\sqrt{1+\tan^2 t}} dt$$

$$= \int \frac{\sin t}{\cos t \sqrt{\frac{1}{\cos^2 t}}} dt = \int \sin t dt$$

$$\stackrel{①}{=} -\cos t + C = -\cos(\arctan x) + C$$

$x = \tan t = u(t)$
 $dx = (1 + \tan^2 t) dt$ ②
 • f cont on $I = \mathbb{R}$ (admits a primitive)
 • u diff on $J =]-\pi/2, \pi/2[$
 • u stimon $\forall u(J) = \mathbb{R} \subseteq I$

$$= -\sqrt{\frac{1}{1+\tan^2(\arctan x)}} + C = -\sqrt{\frac{1}{1+x^2}} + C$$

2nd way:

$$1+x^2 = t \Rightarrow 2x dx = dt$$

$$I = \frac{1}{2} \int \frac{dt}{t\sqrt{t}}$$

$$\stackrel{①}{=} \frac{1}{2} \int t^{-3/2} dt$$

$$\stackrel{②}{=} \frac{1}{2} \frac{t^{-1/2}}{-1/2} + C$$

$$\stackrel{③}{=} -\sqrt{\frac{1}{1+x^2}} + C$$

3rd way:

$$x = \sinh t = u(t) \quad 1+x^2 = \cosh^2 t$$

$$dx = \cosh t dt$$

$$I = \int \frac{\sinh t \cosh t dt}{\cosh^2 t \sqrt{\cosh^2 t}} = \int \frac{d(\cosh t)}{\cosh^2 t}$$

$$\stackrel{①}{=} -\frac{1}{\cosh t} + C = -\frac{1}{\sqrt{1+x^2}} + C$$

Exercise II. Let $\alpha \in \mathbb{R}$. Study the nature of the following series:

1. $\sum_{n=2}^{\infty} \frac{\ln n}{n^4} = \sum_{n=2}^{\infty} \frac{1}{n^4 \ln n}$ *Leibniz, Bertrand* $\alpha = 4 > 1$ (5)

2. $\sum_{n=2}^{\infty} \frac{\alpha^n}{\ln n}$, $\alpha \in \mathbb{R}$

• If $\alpha = 0$, The series is conv. (1)

• If $\alpha \neq 0$, Use Cauchy's Root test:

$$\left| \frac{\alpha^n}{\ln n} \right|^{1/n} = \frac{\alpha}{(\ln n)^{1/n}} \xrightarrow{n \rightarrow \infty} \alpha \quad (1)$$

• If $\alpha < 1$, The series conv (1)

• If $\alpha > 1$, The series div (1)

• If $\alpha = 1$, The test gives no information: (1)
The series $\sum_{n=2}^{\infty} \frac{1}{\ln n}$ div (Bertrand, or Harmonic series) (1B)

3. $\sum_{n=2}^{\infty} (-1)^n \cos n$ div ($\because \lim_{n \rightarrow \infty} (-1)^n \cos n \neq 0$) (5)

4. $\sum_{n=2}^{\infty} \frac{(-1)^n}{n \ln n} \cos \frac{1}{n}$

Abel's Test:

• $\sum \frac{(-1)^n}{n \ln n}$ conv, Alternating series test (1)
 $\left\{ \begin{array}{l} \left(\frac{1}{n \ln n} \right)^{\text{pos}} \text{ dec seq} \\ \frac{1}{n \ln n} \xrightarrow{n \rightarrow \infty} 0 \end{array} \right.$ (1)

• $\cos \frac{1}{n}$ is bdd increasing. (1B)

Then By Abel's test, the series converges.

(n) ↑ pos
(ln n) ↑ pos
↓
(n ln n) ↑ pos
↓
(1/(n ln n)) ↓

2nd way: $f(x) = \frac{\cos \frac{1}{x}}{x \ln x}$, $f'(x) = \frac{\frac{1}{x^2} \sin \frac{1}{x} (x \ln x) - [\ln x + 1] \cos \frac{1}{x}}{(x \ln x)^2}$

$N(x) \xrightarrow{x \rightarrow \infty} -\infty \Rightarrow N(x) \leq 0 \text{ near } \infty \Rightarrow f'(x) \leq 0 \text{ near } \infty$

$\Rightarrow f \downarrow \text{ near } \infty$ (2)

The series conv, Alternating series test (1)
 $\left\{ \begin{array}{l} \left(\frac{\cos \frac{1}{n}}{n \ln n} \right)^{\text{pos}} \text{ dec} \\ \frac{\cos \frac{1}{n}}{n \ln n} \xrightarrow{n \rightarrow \infty} 0 \end{array} \right.$ (1)

Exercise 10. Let $\alpha \in \mathbb{R}$. Study the nature of the following imp integrals:

$$1. \int_0^1 \frac{1 - \cos x}{x^\alpha \sqrt{1-x}} dx = \int_0^{1/2} + \int_{1/2}^1 = I_1 + I_2 \quad (1)$$

near 0^+ : $\frac{1 - \cos x}{x^\alpha \sqrt{1-x}} \sim_{0^+} \frac{x^2}{x^\alpha} = \frac{1}{2} \frac{1}{x^{\alpha-2}} > 0$. I_1 conv iff $\alpha - 2 < 1$ (i.e. $\alpha < 3$) (2)

near 1^- : $\frac{1 - \cos x}{x^\alpha \sqrt{1-x}} \sim_{1^-} (1 - \cos 1) \cdot \frac{1}{(1-x)^{1/2}}$. I_2 conv (Riemann) (2)

Thus I conv iff $\alpha < 3$.

$$2. \int_1^\infty \frac{1}{e^x + e^{-x}} dx.$$

$$\textcircled{1} \frac{1}{e^x + e^{-x}} \leq \frac{1}{e^x} \cdot \int_1^\infty \frac{1}{e^x} dx = -e^{-x} \Big|_1^\infty = -[0 - e^{-1}] = \frac{1}{e} \in \mathbb{R}. \text{ Thus } \int \text{ conv.} \quad \textcircled{1}$$

midway: $x^{10} \cdot \frac{1}{e^x + e^{-x}} \xrightarrow{x \rightarrow \infty} 0$, $\alpha = 10 > 1$, then $\int$ conv (3)

3rd way: $\frac{1}{e^x + e^{-x}} \sim_{\infty} \frac{1}{e^x} > 0$. (2)

$$\int_1^\infty \frac{1}{e^x} dx = -e^{-x} \Big|_1^\infty = -[0 - e^{-1}] = \frac{1}{e} \in \mathbb{R}. \text{ Thus } \int \text{ conv.} \quad \textcircled{1}$$

$$3. K = \int_1^\infty \sin(x^3 + 1) dx = \int_1^\infty \underbrace{\frac{1}{3x^2}}_g \underbrace{3x^2 \sin(x^3 + 1)}_f dx$$

$$\begin{cases} \cdot g \downarrow \textcircled{1}, & g(x) \xrightarrow{x \rightarrow \infty} 0 \end{cases}$$

$$\cdot |F(x)| = \left| \int_x^\infty f(u) du \right| = |-\cos(x^3 + 1) + \cos(2)| \leq 2 \quad (\text{i.e. } F \text{ bdd on } [1, \infty)) \quad \textcircled{1}$$

By Abel-Dirichlet test, K converges (1)

2nd way: $K = \int_1^\infty \frac{1}{3x^2} \underbrace{3x^2 \sin(x^3 + 1)}_{u'} dx$ (1)

$$= -\frac{1}{3x^2} \cdot \cos(x^3 + 1) \Big|_1^\infty - \frac{2}{3} \int_1^\infty \frac{1}{x^3} \cos(x^3 + 1) dx \quad \textcircled{1}$$

$$= \left[0 + \frac{1}{3} \cos 2 \right] - \frac{2}{3} K_1 \quad \textcircled{1}$$

K_1 absol conv by comparison test & Riemann eu, implies K conv. (1)

$$4. L = \int_0^{\infty} \frac{\sin x}{x^{3/2}} dx = \int_0^1 + \int_1^{\infty} = L_1 + L_2 \quad (1)$$

near 0+: $\frac{\sin x}{x^{3/2}} \sim \frac{x}{x^{3/2}} = \frac{1}{x^{1/2}} > 0$. L_1 conv (Riemann $\alpha = 1/2 < 1$)

near ∞ : $|\frac{\sin x}{x^{3/2}}| \leq \frac{1}{x^{3/2}}$. L_2 conv (Comparison test & Riemann $\alpha = 3/2 > 1$)

Thus L converges.

Exercise IV.

1. calculate the following limit:

$$\lim_{n \rightarrow \infty} \left(\prod_{k=1}^n e^{\frac{\sqrt{n^2 - k^2}}{n^2}} \right)^{1/n} = \lim_{n \rightarrow \infty} e^{\frac{1}{n} \ln \left(\prod_{k=1}^n e^{\sqrt{\frac{n^2 - k^2}{n^2}}} \right)}$$

$$\stackrel{(2)}{=} \lim_{n \rightarrow \infty} e^{\frac{1}{n} \sum_{k=1}^n \sqrt{1 - (k/n)^2}} \stackrel{(1)}{=} e^{\int_0^1 \sqrt{1 - x^2} dx}$$

" $f \in C([0,1])$
 $\downarrow$
 $f \in R([0,1])$ "

$$= e^{\int_0^{\pi/2} \sqrt{1 - \sin^2 t} \cos t dt} \stackrel{(1)}{=} e^{\int_0^{\pi/2} \cos^2 t dt}$$

$x = \sin t = u(t)$
 has an inverse f_u

$$\stackrel{(1)}{=} e^{\int_0^{\pi/2} \frac{1 + \cos 2t}{2} dt} = e^{\frac{1}{2} [\pi/2 + \frac{1}{2} \sin 2t]_0^{\pi/2}}$$

$$\stackrel{(1)}{=} e^{\pi/4}$$

2. let $g(x) = \int_0^{\cos x} [\ln(1-t^2)]^2 dt$. compute g' & specify the domain of g .

$$g'(x) = u'(x) f(u(x)) - v'(x) f(v(x)) \quad (2)$$

$$= -\sin x [\ln(1 - \cos^2 x)]^2 = -\sin x (\ln \sin^2 x)^2$$

Domain of diff of g :

①. f cont on $] -1, 1[$ ($\because 1 - t^2 > 0 \Leftrightarrow t \in] -1, 1[$)

②. u, v diff on $\mathbb{R} \cap \mathbb{R} = \mathbb{R} \quad (1)$
 $- v(x) = 0 \in] -1, 1[\Leftrightarrow x \in \mathbb{R} = I_2 \quad (1)$
 $- u(x) = \cos x \in] -1, 1[\Leftrightarrow x \in \bigcup_{k \in \mathbb{Z}}] k\pi, \pi + k\pi[= I_3 \quad (2)$

$\Rightarrow I = I_1 \cap I_2 \cap I_3 = \bigcup_{k \in \mathbb{Z}}] k\pi, \pi + k\pi[\quad (1)$

3. let $f(t) = \begin{cases} \frac{\sin t}{t} & \text{if } t \neq 0 \\ 1 & \text{if } t = 0 \end{cases}$. Calculate $\lim_{n \rightarrow 1} \frac{1}{n-1} \int_0^{\ln n} f(t) dt$.

$$\lim_{n \rightarrow 1} \frac{1}{n-1} \int_0^{\ln n} 1 \cdot f(t) dt \stackrel{(2)}{=} \lim_{n \rightarrow 1} \frac{1}{n-1} f(\zeta_n) \int_0^{\ln n} 1 dt$$

$$\stackrel{(1)}{=} \lim_{n \rightarrow 1} \frac{\ln n}{n-1} f(\zeta_n) \stackrel{(1)}{=} \lim_{n \rightarrow 1} \frac{\ln(1 + [n-1])}{n-1} f(\zeta_n)$$

$$\stackrel{(1)}{=} \lim_{n \rightarrow 1} \frac{n-1}{n-1} f(\zeta_n) \stackrel{(1)}{=} f(0) \quad (\because f \text{ cont at } 0) \\ = 1$$

(1) f cont on $[0, \ln n]$
 or $[\ln n, 0]$ (1)
 • $g(t) = 1 > 0$ on $[0, \ln n]$ (1)
 • $0 < \zeta_n < \ln n$ or $\ln n < \zeta_n < 0$ (1)
 as $n \rightarrow 1, \zeta_n \rightarrow 0$ (1)

Exercise V. Let $I_n = \int_0^1 x (\ln x)^n dx$.

1. Show that the imp int I_n is conv for all $n \in \mathbb{N}$.

I_n converges $\forall n \in \mathbb{N}$, Bertrand $\alpha = -1 < 1$ (2)

2. Show that $I_{n+1} = -\frac{n+1}{2} I_n$ & deduce the value of I_n .

$$I_{n+1} = \int_0^1 \underbrace{x}_{u'} \underbrace{(\ln x)^{n+1}}_v dx$$

$v = (\ln x)^{n+1}$ (2) $u' = x$
 $v' = \frac{n+1}{x} (\ln x)^n$ $u = \frac{x^2}{2}$

$$= \left. \frac{x^2}{2} (\ln x)^{n+1} \right|_0^1 - \frac{n+1}{2} \int_0^1 x (\ln x)^n dx$$

$$= -\frac{n+1}{2} I_n$$

• $I_n = -\frac{n}{2} I_{n-1}$ (1)

$$= \left(-\frac{n}{2}\right) \left(-\frac{(n-1)}{2}\right) \left(-\frac{(n-2)}{2}\right) \dots \left(-\frac{1}{2}\right) I_0; \quad I_0 = \frac{1}{2}$$

$$= (-1)^n \frac{1}{2^{n+1}} n! \quad (1)$$